1 Introduction

In this report we go over the work that we have done since(and before) receiving the car until now.
First of all, team news: we had a team member change immediately after the interview phase. Luckily it was in an early stage so the change went very smoothly!
Overall our activities this sprint was focused on fixing the mechanical problems with the car when we received it, understanding the brain code and trying to add some autonomous features to it.

2 Planned activities

We decided to split our efforts in research and preparation. We planned for 5 main objectives for this phase.
1-Neural Networks : research what kind of NN will we use and which hardware do we need to train and run it, Michele and Davide were responsible for this part.
2-Getting the simulator to run and beginning to test code on it, Panagoda and Banaan were responsbile for this part.
3-The embedded system , looking at which sensors and hardware will we need, Ahmed was responsible for this part.
4-Fixing the mechanical issues of the car and the calibration, since this background information should be known by everyone we decided to work on it together.
5-Getting the car to move using the demo and start testing some autonomous functions. Again everyone worked on this.

3 Status of planned activities

1-Neural Networks: A CNN Network will be used to train on detecting and classifying multiple objects. More details still under review/research.
2-Simulator : The simulator is working but we are facing challenges in getting the camera to work inside it, currently debugging.
3-Embedded system: Postponed until Neural Network capabilities and computational load are better understood
4-Mechanical issues: completed. Many minor things but the biggest one was the steering mechanism was not working as it should due as the servo steering arm was not mounted properly as a screw was missing.
5-Demo and Autonomous functions: Demo completed. The first autonomous function we decided to implement was obstacle avoidance. We started to implement an obstacle avoiding algorithm based on a set of thresholds taking as input the output of the canny edge detector algorithm.
In particular, we obtain the roi (region of interest) from the camera sensor as the lower region of what camera detects. When in this area the number of edge points (obtained via Canny edge detector) is more than a threshold we command the velocity to zero to be able to stop before hitting the obstacle. We have not yet completed the obstacle avoiding task.

4 General status of the project

The car can be controlled manually and the calibration is done. The car can also be controlled autonomously via the camera but still in very early stages

5 Upcoming activities

All of the planned activities need further work to be completed. In particular, we will probably need to adjust thresholding parameters for obstacle avoidance or, more likely, change algorithm and logic technique since the obstacle detection based on Canny edge detector is not robust and does not allow us to understand the size of the obstacle and so the amount of the commanded steering velocity.

When we get the obstacle avoidance to work in a robust way we will implement line following

github: https://github.com/UniPD-DriveOps-BFMC/Brain

Due to mp4 upload error, video here :

https://drive.google.com/file/d/13fGQwMsmpFvwj8llCHsB5QN8_21ai5j1/view?usp=sharing